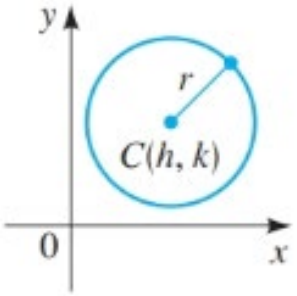
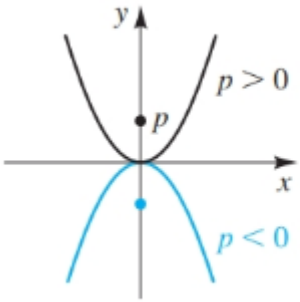
Circles

$$(x - h)^2 + (y - k)^2 = r^2$$



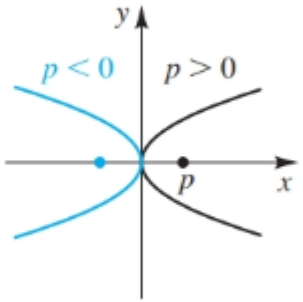
Parabolas

$x^2 = 4py$



Focus $(0, p)$, directrix $y = -p$

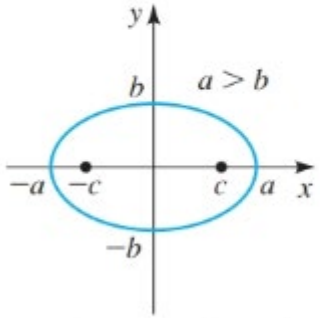
$y^2 = 4px$



Focus $(p, 0)$, directrix $x = -p$

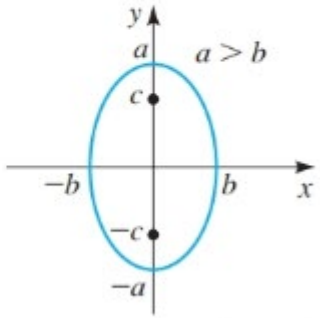
Ellipses

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



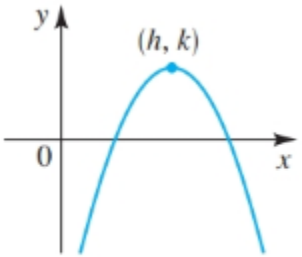
Foci $(\pm c, 0)$, $c^2 = a^2 - b^2$

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

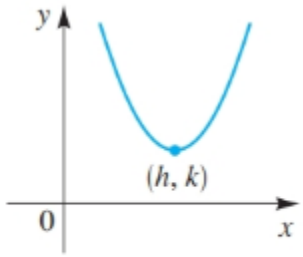


Foci $(0, \pm c)$, $c^2 = a^2 - b^2$

Parabolas



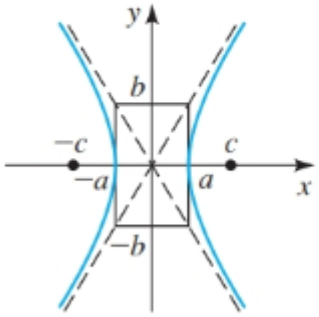
$y = a(x - h)^2 + k,$
$a < 0, \quad h > 0, \quad k > 0$



$y = a(x - h)^2 + k,$
$a > 0, \quad h > 0, \quad k > 0$

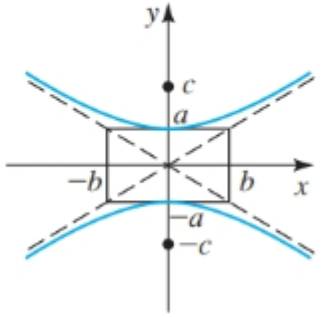
Hyperbolas

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



Foci $(\pm c, 0)$, $c^2 = a^2 + b^2$

$$-\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$



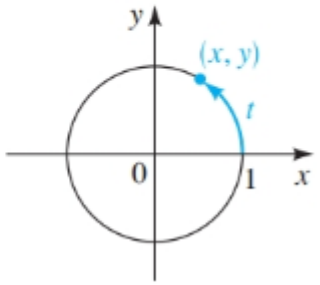
Foci $(0, \pm c)$, $c^2 = a^2 + b^2$

TRIGONOMETRIC FUNCTIONS OF REAL NUMBERS

$$\sin t = y \qquad \csc t = \frac{1}{y}$$

$$\cos t = x \qquad \sec t = \frac{1}{x}$$

$$\tan t = \frac{y}{x} \qquad \cot t = \frac{x}{y}$$

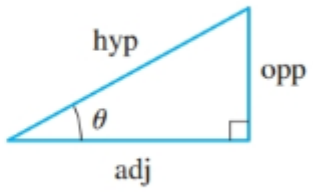


RIGHT ANGLE TRIGONOMETRY

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \csc\theta = \frac{\text{hyp}}{\text{opp}}$$

$$\cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \sec\theta = \frac{\text{hyp}}{\text{adj}}$$

$$\tan\theta = \frac{\text{opp}}{\text{adj}} \qquad \cot\theta = \frac{\text{adj}}{\text{opp}}$$



SPECIAL VALUES OF THE TRIGONOMETRIC FUNCTIONS

θ	radians	$\sin\theta$	$\cos\theta$	$\tan\theta$
0°	0	0	1	0
30°	$\pi/6$	$1/2$	$\sqrt{3}/2$	$\sqrt{3}/3$
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90°	$\pi/2$	1	0	—
180°	π	0	-1	0
270°	$3\pi/2$	-1	0	—

THE LAWS OF SINES AND COSINES

The Law of Sines

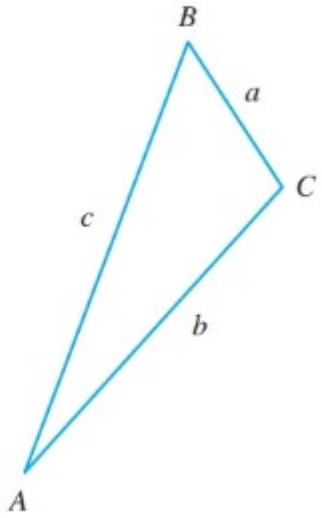
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

The Law of Cosines

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

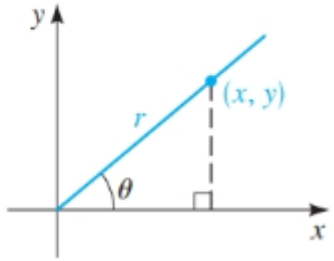


TRIGONOMETRIC FUNCTIONS OF ANGLES

$$\sin\theta = \frac{y}{r} \qquad \csc\theta = \frac{r}{y}$$

$$\cos\theta = \frac{x}{r} \qquad \sec\theta = \frac{r}{x}$$

$$\tan\theta = \frac{y}{x} \qquad \cot\theta = \frac{x}{y}$$



Reciprocal Identities

$$\csc x = \frac{1}{\sin x} \qquad \sec x = \frac{1}{\cos x}$$

$$\tan x = \frac{\sin x}{\cos x} = \frac{1}{\cot x} \qquad \cot x = \frac{\cos x}{\sin x} = \frac{1}{\tan x}$$

Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Half-Angle (Power Reducing) Identities

$$\cos^2 x = \frac{1+\cos(2x)}{2}$$

$$\sin^2 x = \frac{1-\cos(2x)}{2}$$

Double-Angle Identities

$$\sin(2x) = 2\sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$

$$= 2\cos^2 x - 1 = 1 - 2\sin^2 x$$

$$\tan(2x) = \frac{2\tan x}{1-\tan^2 x}$$

Sum and Difference Identities

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$$

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

$$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

$$\tan(\alpha+\beta) = \frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta} \qquad \tan(\alpha-\beta) = \frac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta}$$

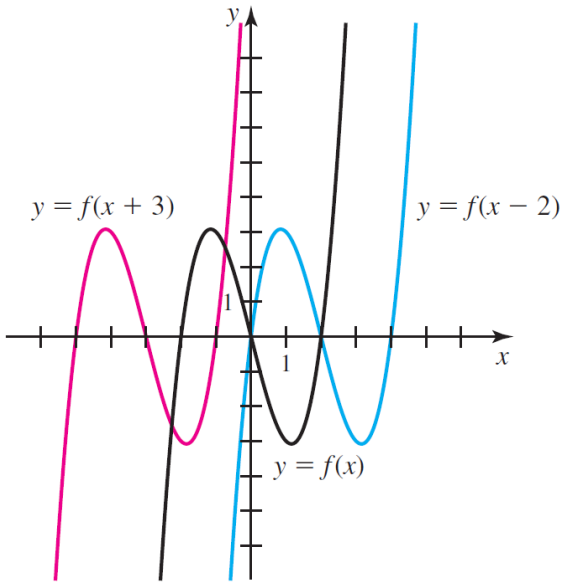
Product to Sum-Identities

$$\sin x\cos y = \frac{1}{2}\left[\sin(x+y)+\sin(x-y)\right]$$

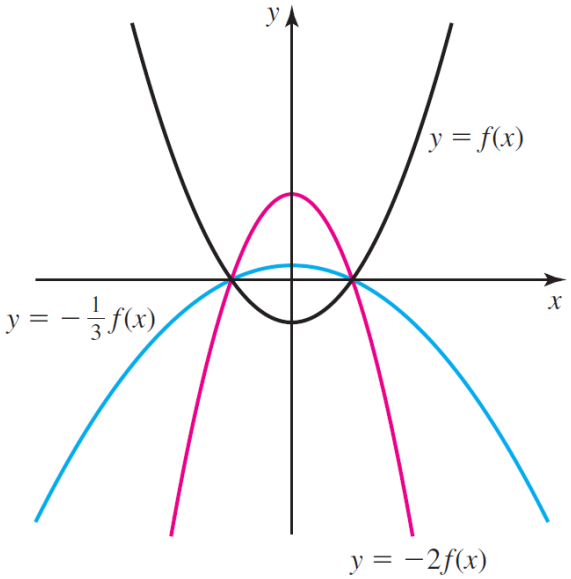
$$\cos x\cos y = \frac{1}{2}\left[\cos(x+y)+\cos(x-y)\right]$$

$$\sin x\sin y = \frac{1}{2}\left[\cos(x-y)-\cos(x+y)\right]$$

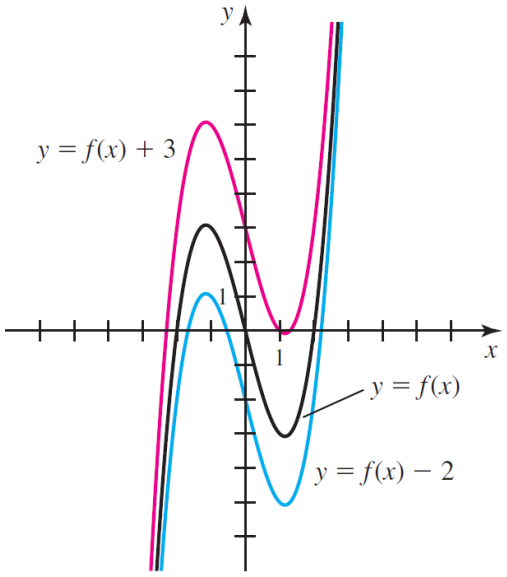
The graph of $y = f(x - b)$ is the graph of $y = f(x)$ shifted horizontally by b units (right if $b > 0$ and left if $b < 0$).



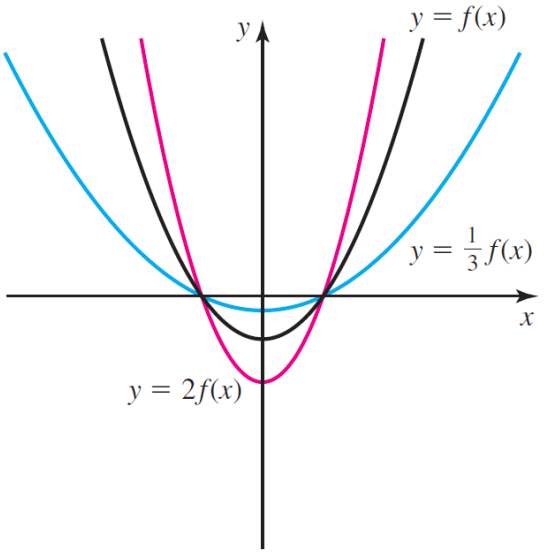
For $c < 0$, the graph of $y = cf(x)$ is the graph of $y = f(x)$ scaled vertically by a factor of $|c|$ and reflected across the x-axis (broadened if $-1 < c < 0$ and steepened if $c < -1$).



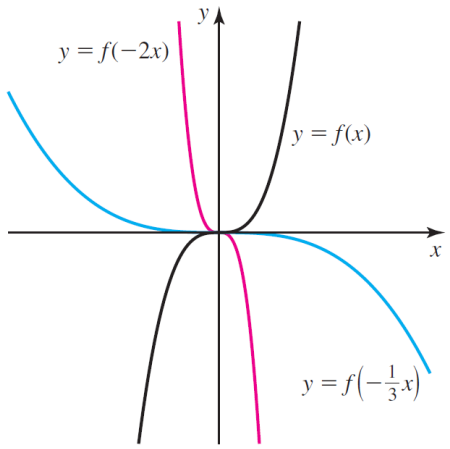
The graph of $y = f(x) + d$ is the graph of $y = f(x)$ shifted vertically by d units (up if $d > 0$ and down if $d < 0$).



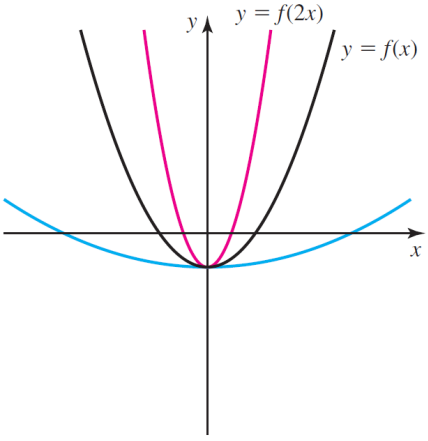
For $c > 0$, the graph of $y = cf(x)$ is the graph of $y = f(x)$ scaled vertically by a factor of c (broadened if $0 < c < 1$ and steepened if $c > 1$).

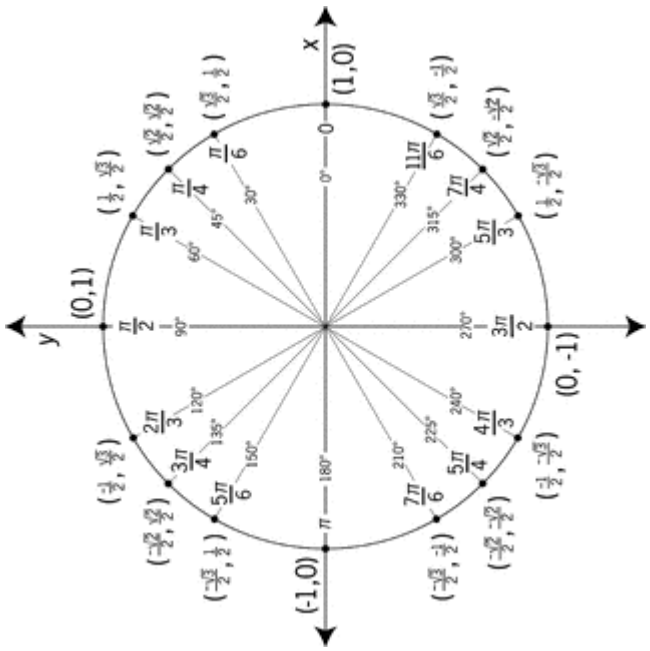


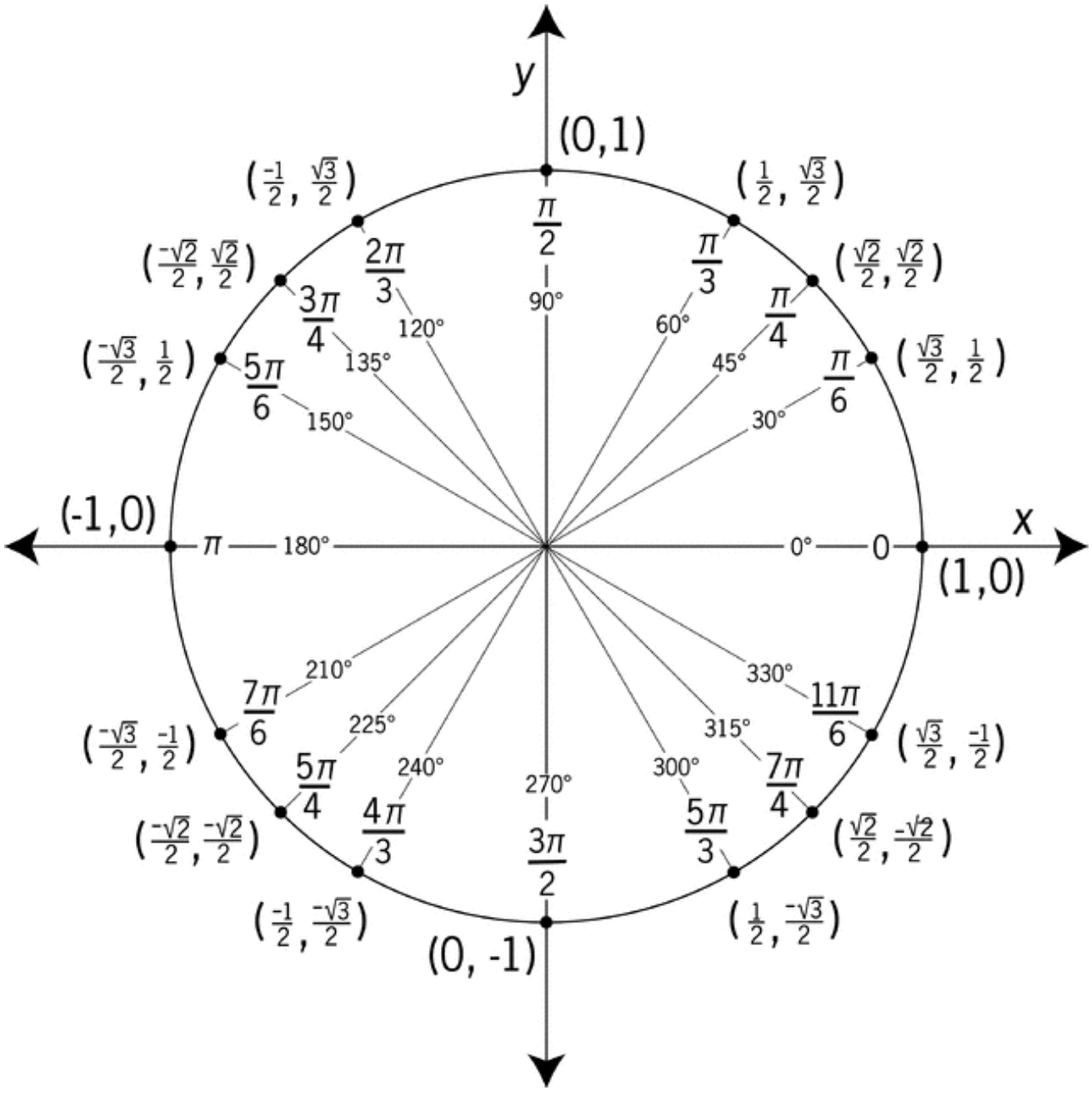
For $a < 0$, the graph of $y = f(ax)$ is the graph of $y = f(x)$ scaled horizontally by a factor of $|a|$ and reflected across the y-axis (broadened if $-1 < a < 0$ and steepened if $a < -1$).



For $a > 0$, the graph of $y = f(ax)$ is the graph of $y = f(x)$ scaled horizontally by a factor of a (broadened if $0 < a < 1$ and steepened if $a > 1$).






y
x
(0,1)
(1,0)
(-1,0)
(0, -1)
$\frac{\pi}{2}$
90°
$\frac{\pi}{3}$
60°
$\frac{\pi}{4}$
45°
$\frac{\pi}{6}$
30°
π
180°
0°
0
$\frac{2\pi}{3}$
120°
$\frac{3\pi}{4}$
135°
$\frac{5\pi}{6}$
150°
$\frac{7\pi}{6}$
210°
$\frac{5\pi}{4}$
225°
$\frac{4\pi}{3}$
240°
$\frac{3\pi}{2}$
270°
$\frac{5\pi}{3}$
300°
$\frac{7\pi}{4}$
315°
$\frac{11\pi}{6}$
330°
$(\frac{1}{2}, \frac{\sqrt{3}}{2})$
$(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$
$(\frac{\sqrt{3}}{2}, \frac{1}{2})$
$(\frac{-1}{2}, \frac{\sqrt{3}}{2})$
$(\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$
$(\frac{-\sqrt{3}}{2}, \frac{1}{2})$
$(\frac{-\sqrt{3}}{2}, \frac{-1}{2})$
$(\frac{-\sqrt{2}}{2}, \frac{-\sqrt{2}}{2})$
$(\frac{-1}{2}, \frac{-\sqrt{3}}{2})$
$(\frac{1}{2}, \frac{-\sqrt{3}}{2})$
$(\frac{\sqrt{2}}{2}, \frac{-\sqrt{2}}{2})$
$(\frac{\sqrt{3}}{2}, \frac{-1}{2})$